Neurotech Hackathon: Team 20 Reference (NeuroPawn Orange Track)

Nov 7, 2026 · @Mahek

Summary: what we know and possible directions

No project is chosen yet; this section collects the constraints and the directions still in play for the team brainstorm.

Team priority from brainstorm: weight scalability and inclusivity heavily when comparing directions, not just the four main criteria.

What we know:

- **Rubric:** best application of the NeuroPawn headset, judged on creativity, practicality, technical difficulty and execution, plus special subgoals.
- **Subgoals:** scalability (NC State Entrepreneurship: a project that could continue after the hackathon) and inclusivity (Synaptic, $100 per track).
- **Prizes:** 1st–3rd place among ~10 orange teams.
- **Already done, avoid:** mind-controlled cars, cursors, music apps (named by organizers), plus everything on NeuroPawn's page (SSVEP speller, focus game, P300 music game, motor imagery vehicle).
- **Demo format:** live demo not required; video and slides allowed.
- **Hardware:** 8 probes on a 10-10 headset, no IMU, EOG possible, not suited to source localization.
- **Paradigms mentioned by organizers:** motor imagery, SSVEP, P300. Unclear whether passive BCIs count.

Possible directions:

Direction	Core idea	Strengths	Risks / unknowns
Cognitive-decline screener	5-min eyes-closed EEG flags slowed rhythms linked to Alzheimer's / frontotemporal dementia; trained on OpenNeuro ds004504 (88 people, 19 channels, includes all our candidate positions)	Disease prevention angle; offline ML is low execution risk; inclusivity (no language or education bias); scalable screening	Patients already diagnosed, so not proof of early detection; can't show detection live; clinical vs. NeuroPawn hardware differences
"What did I miss?" lecture recap	Detects lecture lapses (EEG + EOG + webcam), LLM recap and quiz	Relatable to every judge; multimodal; inclusivity (ADHD, multilingual, captions)	Many live parts; passive-BCI eligibility unclear; similar research exists (cite it)
Interview / presentation stress coach	Heart rate + EEG aligned with speech transcript, LLM feedback	Only idea using NeuroPawn's ECG mode; dramatic demo	Needs ECG electrodes and spare channels; EEG stress is fuzzy
P300 recognition "card trick"	Detects which card a judge secretly picked	Highest wow factor; uses a listed paradigm	Weak practicality and scalability; timing-sensitive
Wellness / burnout trend	Tracks attention or resting rhythms over weeks to suggest rest or support	Prevention framing without medical claims	Hard to show over a weekend

Questions that decide between directions:

- Do passive BCIs and resting-state screening count, or only the listed paradigms?
- Do reference and ground use any of the 8 probes?
- What are the "special" judging criteria beyond the two subgoals?
- Are ECG electrodes available?

Useful regardless of direction: BrainFlow streaming + filtering, band-power features, subject-wise validation, a clean dashboard, a backup demo video, and citing prior work on a slide.

Event and team

Team 20 competes in the orange track: an open, creative challenge sponsored by NeuroPawn, judged on the most creative application of their headset. About 10 teams compete in orange.

- **Prizes:** 1st, 2nd and 3rd place. Synaptic also gives $100 to one team per track for inclusivity (definition not yet confirmed).
- **Judges:** NeuroPawn founder Abdallah Alwan is attending and likely judges or shapes the criteria. Nivesh Sharma (credited as a guide by a previous NC State winner) may also be involved.
- **Host:** Neurotech at NC State, which also ran the earlier BR41N.IO event where several NeuroPawn-featured projects were built.
- **Blue track (not ours):** a timed ANT Neuro challenge on the brain under movement.

Teammate	School / major	Tentative role
Mahek Kantharia	NC State, CSC	EEG pipeline, classifier, headline metric
Lohith Senthil Kumar	NC State, CSC	App, dashboard, integration
Eric Sthole	NC State, BME	Electrode placement, signal checks, neuro sanity checks
Tenne Tian	Not confirmed	Not attending
Marques Washington	Rowan-Cabarrus CC, Criminal Justice Technology (team number not confirmed)	Demo, pitch, backup video

Team diversity (three schools, CS / BME / criminal justice) is a pitch point, especially for the inclusivity prize.

Hardware and setup

We have an 8-channel NeuroPawn Knight Board on a 10-10 headset, with no IMU. It can also record EOG (eye movements). It is not suited to source localization, which our plans don't need.

Confirmed: no IMU for our team. Some boards have one; ours doesn't. Webcam head pose (MediaPipe) replaces it.

Recommended 8-channel layout (covers the lecture recap, P300 and SSVEP directions without moving electrodes):

Electrode	Purpose
Fp1	Blinks; rough vertical EOG
Fz	Frontal theta (mental workload / effort)
Cz	P300 backup (swap for a horizontal EOG electrode beside one eye if eye-movement data matters more)
Pz	Strongest P300 site; alpha
P3, P4	Alpha, left and right
O1, O2	Strongest alpha and SSVEP

10-10 naming: Fp = forehead, F = frontal, C = central, P = parietal, O = occipital; odd = left, even = right, z = midline.

Reference and ground (bias / DRL): earlobes or mastoids. Keep them off the forehead (would interfere with Fz, Fp1) and away from jaw muscles. If every channel goes noisy at once, check the reference first. Confirm the board's labels with NeuroPawn docs; their posts mention individually addressable references.

EOG: the eyeball acts like a small battery (cornea positive, retina negative). Electrodes beside the eyes catch left/right movement; above/below catch up/down and blinks. Detects blinks, saccades, rough gaze direction, and slow rolling eye movements (a drowsiness sign).

Software: use NeuroPawn's EXG Visualizer for signal checks (contact, blinks, 60 Hz hum). Build a custom dashboard (Streamlit, optionally PyQtGraph for live plots) for the project. Only one program can usually connect to the board at a time: use NeuroPawn's LSL streamer so several programs can subscribe, or close the visualizer before running our script.

Signal sanity tests, in order:

1. Blink hard: big spikes on Fp1.
2. Clench teeth: burst of muscle noise.
3. Eyes closed 10 s, then open: 8–12 Hz alpha appears at O1/O2/Pz while closed.
4. Mental arithmetic (count back from 1000 by 7s), then rest: slightly higher theta at Fz during math. This is the subtle signal the lecture recap relies on.

Other quick tests: look left/right (horizontal EOG), electrodes on chest or wrists (ECG heartbeat), forearm + fist (EMG), open/close right hand (mu drop at C3).

NeuroPawn landscape

NeuroPawn (founded 2023, ~314 LinkedIn followers) sells an affordable 8-channel EEG board and has featured nine projects; SSVEP control, music and vehicles are crowded, while physical-world interaction, ECG and the firmware package are untouched.

Featured projects (avoid copying):

Project	Where	Signal / method	What it does
SSVEP T9 speller	Hack the North	SSVEP on 4 letter groups (a–f, g–l, m–r, s–z), 12/15 Hz on a 60 Hz screen, trie + local LM	Types words; open problem: ambiguous words (rare vs. mane)
Toy car (won)	NC State hackathon	SSVEP	Mind-controlled car
Head mouse	BR41N.IO at NC State	Eyebrow raises (random forest) + IMU	Hands-free cursor and clicks
CogniShift (3rd)	BR41N.IO at NC State	EEG + music	Music and brain-state link; best concept grade, but last-minute database failure hurt them
Web app control	Bachelor project	SSVEP + CCA, Python backend	Radio station switching, emergency signal
FocusFarm	BrainHack Lucca 2025	Alpha band power (PSD) z-scored against a personal baseline	Plants grow as focus score rises
Blob Bop	BCI Game Jam 2026	P300	Kids pick sounds for 4 characters who then sing
EEG sonification	Brown music thesis	EEG features mapped to sound	Adaptive film scoring (film Bliss Place)
NeuroLink	Graduation project	Motor imagery + EEGNet + calibration	Robot vehicle, 5 commands, "semi-real-time" after a full year

Product launches and updates:

Update	What it is	Showcased by a project?
Knight Board with IMU	9-axis motion sensor for head tracking and motion-noise suppression	Only the head mouse (cursor only)
Firmware package (PlatformIO / Arduino)	Run custom code on the board's spare microcontroller pins	None
Tutorials	Starter code for P300, SSVEP, motor imagery	Use for prep
LSL streaming script	Stream board data to other apps	Not needed
EXG Visualizer update	ECG/HRV mode, IMU viewer, fixed SSVEP stimulus timing, disconnect button	ECG mode unused

Patterns to use in the pitch:

- Accessibility framing appears in almost every post.
- SSVEP is proven on this board but no longer novel on its own.
- Newest features have almost no showcases; projects using them give NeuroPawn something new to post.
- Execution counts: CogniShift lost ground to a broken demo despite the best concept.

Also ruled out: a friend competing at this hackathon built object detection + on-screen options + eye-tracking selection for patients and may rebuild it, so object-selection concepts are off-limits (object detection as a component is fine).

Ideas evaluated

The "What did I miss?" lecture recap survived every check; most other ideas were already done, impractical, or too risky for a weekend.

Idea	Status	Reason
"What did I miss?" lecture recap	**Candidate**	Not on NeuroPawn's page or built by the friend; works without the IMU; detection errors are harmless; every judge relates to it

Idea	Status	Reason
P300 recognition "card trick"	**Candidate**	Highest wow factor; nobody on NeuroPawn's page did recognition; weak real-world use
Interview / presentation stress coach (EEG + ECG + transcript)	**Candidate**	Only idea using the unused ECG mode; needs ECG electrodes confirmed
SSVEP + LLM speller	Dropped	Hack the North built it and NeuroPawn featured it; MindChat (2025) published it
SSVEP robot / car controller	Dropped	Toy car already won at NC State; NeuroLink did a vehicle too
Object selection (camera + flicker or gaze)	Dropped	Friend's project; eye tracking beats SSVEP for selection anyway
LEDs on real objects driven by firmware pins	Dropped	Impractical (nobody tapes LEDs to everyday objects); the fixed-appliance version duplicates the bachelor project
Focus game / focus tracker alone	Dropped	FocusFarm; EEG-only focus is unreliable
Music from brain waves	Dropped	CogniShift and the Brown sonification thesis
Motor imagery anything	Dropped	Needs training and calibration; NeuroLink took a year
Drowsy driving alert	Dropped	Heavily researched, not novel
Reading-struggle detector (webcam gaze)	Dropped	Webcam gaze too imprecise to know which line you're reading

How we pick between ideas (path C): a proven technique applied to a new problem. The technical core has published evidence it works; the application has no prior example. Most hackathon winners are this type.

Scoring filter for any idea: works live on 8 channels, has a named real user, shows off the headset, gives a wow in 10 seconds, produces a number for a slide.

"Why EEG?" answer for judges: "The webcam sees where you're looking; the EEG tells us whether you're actually paying attention." Cameras already beat EEG for blinks, eyebrow raises and pointing at things on screen.

Direction: "What did I miss?" lecture recap

A passive BCI that notices when a student zones out during a lecture and gives a short LLM recap and quiz of exactly what was said during those moments.

One-line pitch: "It notices when you zone out in lecture and tells you what you missed."

Signals, each doing what it does best:

Signal	Source	Detects
Alpha power (8–12 Hz)	O1, O2, Pz, P3, P4	Relaxed / drifting state
Frontal theta (4–8 Hz)	Fz	Mental effort; a drop while eyes stay on screen = "looking but not processing"
EOG	Fp1 (+ optional horizontal electrode)	Blinks, saccades (active reading vs. blank stare), slow rolling eye movements (drowsiness)
Head pose	Webcam + MediaPipe (replaces IMU)	Head down, drooping
Phone in hand, absent from frame	Webcam + YOLO	Distraction, left the room
Lecture content	Microphone or video + Whisper	Timestamped transcript

Pipeline:

1. Stream EEG via BrainFlow (or LSL); bandpass + 60 Hz notch filter.
2. Every second: alpha and theta band power, or Riemannian covariance features (pyRiemann) with a linear classifier.
3. Use camera blink detection and Fp1 spikes to mark blink windows and ignore them in the EEG features.
4. Camera branch: MediaPipe head pose + YOLO phone detection with thresholds; no training.
5. Late fusion: flag a lapse if the camera sees distraction, OR if the EEG lapse score stays high for 10+ seconds while the person faces the screen.
6. Map flagged time ranges to transcript segments; LLM writes the recap and 2–3 quiz questions.
7. Streamlit dashboard: live lapse timeline, camera status, recap at the end.

Calibration (essential): 2 minutes per user, relaxed vs. focused (e.g., mental arithmetic). Use a rolling baseline over the last several minutes, because alpha naturally drifts with time-on-task (Benwell, London et al.).

Why it's robust: a false alarm only adds an extra note to the recap. Late fusion means the camera branch still works if EEG is noisy.

Measured result to show: use YOLO "phone in hand" moments as automatic labels for distraction. Report EEG-alone detection rate vs. EEG + camera.

Prior work (don't claim "first"; cite it):

- NCCU attention-aware system: NeuroSky EEG linked to lecture video tagging; low-attention periods correlated with lower test scores.
- Attention-based Zoom summaries (2021): webcam facial attention builds personalized video summaries of low-attention moments.
- Eye-tracking video reconstruction and replay: highlights and replays missed lecture content.

Our differentiators: EEG + webcam fusion (each alone misses things), LLM text recap + quiz instead of video clips, live in-person lectures with real-time transcription, 8-channel affordable board with per-user calibration.

Pitch structure: "Research has shown lecture attention lapses can be detected with EEG, webcams or eye trackers, but each alone misses things, and none tell you what you missed in a form you can use. We fuse brain and camera signals and turn your lapses into a 30-second recap and quiz." Frame as a passive BCI: "Most BCI projects make you control something. Ours watches out for you while you learn."

Inclusivity angle (Synaptic prize): ADHD and neurodivergent students (most lapses, most benefit); multilingual recap toggle for non-native speakers (one extra prompt line); live transcript doubles as captions for deaf / hard-of-hearing students.

Optional reactive add-on: a thin, low-contrast border flickering at ~12 Hz around the lecture window (frequency tagging). The 12 Hz SSVEP peak fades when the student looks away. Test on the real headset; add an epilepsy warning.

Demo script: calibrated teammate watches a 3-minute lecture clip, solves a math aside, picks up a phone and scrolls, then stares blankly. The dashboard flags each moment; the recap shows exactly what was said while scrolling.

Other directions

If passive BCIs aren't allowed, switch to the P300 card trick; if ECG electrodes are available and the rubric favors drama, the stress coach is the alternative.

Backup 1: P300 recognition "card trick"

- A judge secretly picks one of 6 cards. Cards flash one at a time, repeated 10–20 times. The system averages the P300 per card and names the chosen one.
- Uses Pz, Cz, Fz. Start from NeuroPawn's P300 tutorial. Stimulus timing precision matters.
- Pitch as recognition detection, not "lie detection." Mention ethics (consent, mental privacy) and that it can be defeated by countermeasures.
- Strength: highest wow factor. Weakness: entertainment more than a product; 30–60 s of flashing per guess.

Backup 2: interview / presentation stress coach

- Rehearse a talk or mock interview wearing the headset. Heart-rate spikes (ECG) and EEG tension are aligned with the speech transcript; an LLM gives feedback (e.g., heart rate jumped on the "weaknesses" question, 9 filler words in that answer).
- Requires ECG electrodes on chest or wrists using 2 of the 8 channels; confirm availability. Use plain heart rate, not HRV (NeuroPawn warns HRV needs long recordings). Fallback: smartwatch heart rate.
- Only idea using NeuroPawn's unused ECG mode.

Backup 3: SSVEP (only if required by the rules)

- 4 targets at 60 Hz-compatible frequencies (7.5, 8.57, 10, 12, 15 Hz); CCA classifier; O1, O2, Pz.
- Novelty must come from the application, not the technique. Add an epilepsy warning.
- Speller upgrade idea if ever needed: fuse CCA confidence scores with LLM word probabilities (Bayesian) to decide faster and resolve ambiguous words.

EEG primer

EEG records microvolt-level voltages at the scalp, 250–1000 samples per second per channel; noise from blinks, muscles and 60 Hz power lines is often larger than the brain signal.

Three types of BCI:

Type	What the user does	Examples	Purpose
Active (endogenous)	Deliberate mental task	Motor imagery	Control

Type	What the user does	Examples	Purpose
Reactive (exogenous)	Attends to a stimulus	SSVEP, P300	Control
Passive	Nothing special	Focus, workload, drowsiness monitoring	Monitor and adapt

Frequency bands:

Band	Range	Associated with
Theta	4–8 Hz	Frontal-midline theta rises with mental effort; also drowsiness
Alpha	8–12 Hz	Rises when relaxed or eyes closed; drops with active concentration
Mu	8–13 Hz over motor cortex	Drops when moving or imagining movement
Beta	13–30 Hz	Alertness; motor activity

Paradigms:

- **SSVEP:** each flash makes visual cortex neurons respond; steady flashes produce a steady rhythm at the flicker frequency and its harmonics (12 Hz flicker gives peaks at 12, 24, 36 Hz). Works for nearly everyone, no training. Monitors refresh at 60 Hz, so flicker frequencies should divide into it. Flicker at 5–30 Hz can trigger photosensitive epilepsy.
- **P300:** a positive spike ~300 ms after a rare, meaningful event. Strongest at Pz. Needs averaging over repeats.
- **Motor imagery:** imagining a hand movement weakens mu/beta over the opposite side of the motor cortex (left hand affects C4, right hand C3). Hard: small signal, needs training; a notable share of people can't produce reliable patterns ("BCI inefficiency").
- **ErrP:** brain signal when you notice an error. Single-trial detection is hard.

Key terms:

- **Epoch:** a short window of data locked to a stimulus or time point.
- **CCA:** canonical correlation analysis; standard training-free SSVEP classifier that matches EEG to each target's frequency.
- **ITR:** information transfer rate in bits per minute; standard BCI speed metric.

- **Riemannian geometry:** compute the 8×8 channel covariance matrix per window (captures power and channel relationships). These matrices live on a curved surface, so distances and averages must follow the curve. TangentSpace flattens it locally so a normal classifier works. Works with little data, handles cross-session shifts via re-centering, fast. In pyRiemann: Covariances → TangentSpace → LogisticRegression.
- **Source localization:** estimating where in the brain a signal originates; needs 64+ channels and a head model. Not needed for scalp-level BCIs.
- **Late fusion:** separate models per signal, combined at the decision level.

Why people differ so much: skull thickness, cortical folding, electrode position, contact quality (hair, sweat), individual alpha peak (8–13 Hz) and amplitude, mental strategy, and state (sleep, caffeine, stress). Fixes: per-user calibration, normalizing to a personal baseline (z-score), Riemannian re-centering, transfer learning.

What EEG adds over video: workload (hard thinking and blank staring look the same on camera), early drowsiness (brain changes before eyelids droop), looking-but-not-processing (weak, treat as a hint), recognition (P300), noticing errors (ErrP), imagined movement. Video wins for blinks, eyebrow raises, head pose and screen selection (eye tracking is faster and supports many targets; commercial ALS devices use eye gaze, not EEG).

Literature and state of the art

The most relevant paper detected mind-wandering during video-based learning with an 8-channel EEG: mean AUC 0.876 within the same person, about 0.69–0.70 on a different lecture.

Task	State of the art	Feasible on our board this weekend	Key sources
Mind-wandering	Within-person ~75–90%, worse across people; labels need thought probes	Weak; treat as a hint	Tang, Liang & Li 2023, 8-ch video learning; 2025 study, ~76% within / ~72% cross; 2026 multimodal education benchmark, 14 datasets

Task	State of the art	Feasible on our board this weekend	Key sources
Mental workload	Frontal-midline theta rises with difficulty; EEGMeNet ~87.5% cross-subject	Moderate, with baseline	Frontiers Neuroergonomics 2025 review; EEGMeNet; deep learning workload review: shuffling before splitting inflates accuracy
Drowsiness	~83.5% on SEED-VIG; real systems combine EEG with eye/camera	Moderate, best with camera	arXiv 2512.21747; EEG drowsiness features review
Recognition (P300 CIT)	Meta-analytic effect size 1.89; 85–100% in lab; defeatable by countermeasures	Yes, with designed stimulus	Meta-analysis; Polich 2019; countermeasure study
Error potentials	Single-trial detection still hard; shrinkage LDA most robust	Hard	ErrP survey
Motor imagery	EEG foundation models (LaBraM, CBraMod) hyped, but a Sept 2026 audit found supervised baselines beat them on 4-class MI	Hard	CBraMod; EEG-FM-Bench; 2026 audit
SSVEP decoding	CCA (Lin 2006), filter-bank CCA (Chen 2015), TRCA (Nakanishi 2018) still baselines; deep learning needs per-user data	Yes, very reliable	SSVEP review, Brain Science Advances; Deep learning SSVEP for mobile BCIs, 2026
LLM-assisted spellers	MindChat (2025): GPT-4o completions, first LLM SSVEP speller; CMU 2024 char-level LM + EEGNet; P300 + LLM word prediction	Done; avoid	MindChat; CMU 2024; P300 + LLM

Prior lecture-attention systems: NCCU EEG attention-aware lecture tagging; Attention-based Zoom summaries; Eye-tracking lecture replay; PAFE webcam mind-wandering, KAIST.

Patterns across tasks: within-person works, across-person struggles (calibrate each user); simple models often win on small noisy data; multimodal fusion is where the field is heading.

Other resources: Raquel London (Ghent University EEG researcher, The EEG Newsletter) has a YouTube channel of 2024 Ghent EEG methods seminars; watch at most one talk before the event (preprocessing or decoding). Her co-authored paper on alpha frequency and power drifting with time-on-task motivates the rolling baseline. Mike X Cohen's YouTube channel is the classic EEG analysis intro.

Datasets and tools

BBBD is the best dataset for the lecture recap: 178 people watching educational videos with synchronized EEG, ECG, eye tracking and head movement, plus memory quizzes.

Datasets:

Dataset	Contents	Use
BBBD (2026)	178 participants, ~110 h, educational videos; EEG, EOG, ECG, eye tracking, head movement, quizzes, ADHD self-reports; code at github.com/madjens/bbbd-dataset	Check whether flagged lapses match missed quiz questions; ADHD angle
MUTLA (2019)	Webcam video + consumer EEG + learning logs, students solving problems	EEG + webcam reference
PAFE (KAIST 2022)	15 h webcam video in online lectures, ~1,100 mind-wandering probes; no EEG	Camera-side reference
Nakanishi 2015 (via MOABB)	10 subjects, 12 SSVEP targets, 8 channels	SSVEP backup testing
Kalunga 2016 (via MOABB)	12 subjects, 3 frequencies + rest, 8 channels	SSVEP on lower-cost hardware
Tsinghua Benchmark (Wang 2016)	35 subjects, 40 targets, 64 channels	Standard speller benchmark

Dataset	Contents	Use
BETA (Liu 2020)	70 subjects, 40 targets, recorded outside lab	Noisy, realistic SSVEP
SEED-VIG	Drowsiness benchmark	Drowsiness reference

MOABB dataset details are from memory; confirm when downloading.

Libraries:

Tool	Use
BrainFlow	Stream from the board; synthetic board for development without hardware
NeuroPawn tutorials (P300, SSVEP, motor imagery)	Starter code matched to the hardware
NeuroPawn LSL script / pylsl	Share the stream across programs
MNE-Python	Load, filter, plot, epoch EEG
pyRiemann	Covariances + TangentSpace + classifier
MOABB	Download standard BCI datasets
MediaPipe	Head pose, face landmarks, blinks (eye aspect ratio)
YOLO	Phone-in-hand, person-present detection
Whisper	Timestamped transcript
LLM API	Recap and quiz generation
Streamlit / PyQtGraph	Dashboard / smooth live plots

Open questions, prep and pitch

The deciding open question is whether passive BCIs are allowed; the organizers listed only motor imagery, SSVEP and P300.

Still to ask the organizers or founder:

- [ ] Do passive BCIs (attention / mental-state monitoring) count?
- [ ] Judging rubric, and how Synaptic defines inclusivity
- [ ] Wet or dry electrodes; reference and ground placement

- [ ] Outside tools allowed: webcam, LLM APIs, public datasets, code written before the event
- [ ] Demo length; live or video
- [ ] ECG electrodes available? (Backup 2 only)
- [ ] BrainFlow board ID for the Knight Board; sampling rate
- [ ] Confirm Marques is on Team 20

Already answered: no IMU for our team; 10-10 headset; EOG supported; custom visualizer allowed; not suited to source localization.

Rubric (announced at kickoff): track goal is best application of the NeuroPawn headset, judged on creativity, practicality, technical difficulty and execution, plus special subgoals. Subgoal 1, scalability (NC State Entrepreneurship partnership): a project that could be scaled or continued after the hackathon. Subgoal 2: inclusivity. Organizers named mind-controlled cars, cursors and music apps as last time's projects, so avoid those.

Scalability pitch for the lecture recap: a webcam-only free tier needs no hardware; the EEG tier serves accessibility offices and students with ADHD; the recap engine plugs into lecture-capture platforms universities already use; continuation path is a pilot with NC State Disability Resource Office or a course, then cheaper headbands as they arrive.

Prep checklist:

- [ ] BrainFlow on the synthetic board: stream, bandpass + notch, alpha and theta power each second
- [ ] MediaPipe head pose "head down" flag + YOLO phone detection on the webcam
- [ ] Whisper timestamped transcript of a short lecture clip
- [ ] Glue script: missed time ranges → transcript segments → LLM recap + quiz
- [ ] Streamlit dashboard skeleton
- [ ] Read Tang, Liang & Li 2023 (8-channel mind-wandering in video learning)
- [ ] Stretch: alpha/theta lapse score on a few BBBD participants vs. their quiz misses
- [ ] Run NeuroPawn's P300 tutorial on the synthetic board (Backup 1)

On the day:

- Talk to the founder in the first hour with a specific question; show him early progress midway.
- Write the rubric down word for word at kickoff; build the final slide around it.
- Keep the stack simple: no databases or cloud dependencies (CogniShift lesson).
- Record a 90-second backup demo video halfway through.
- Keep the idea within the team until the event starts.

- After: connect with the founder on LinkedIn the same day; post a demo video tagging NeuroPawn.

Odds (rough estimates, not data): with ~10 teams and 3 placing prizes, an equal team has ~30% to place. With a working pipeline and reliable demo, roughly 40–55% to place and 20–30% for first.